

Machine translation:

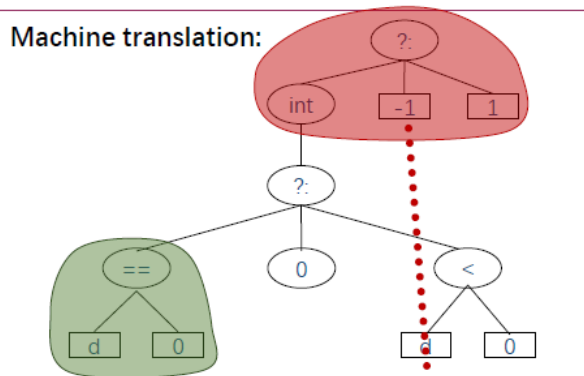
```
public static int Sign ( double d )
{
    return ((int)((d == 0)? 0:(d < 0)))?
    -1: 1;
}
```

Reference (human) translation:

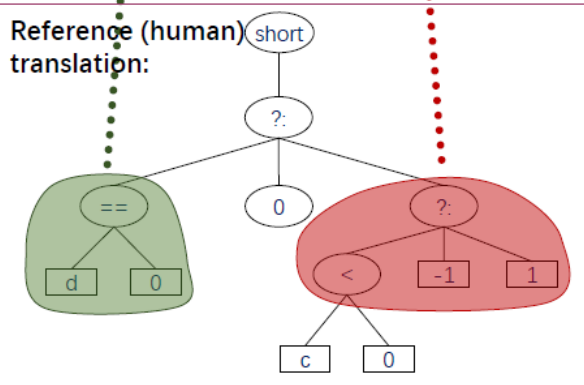
```
public static short Sign (double d )
{
    return (short)((d == 0)? 0:(c < 0)?
    -1: 1);
}
```

Weighted N-Gram Match

Machine translation:



Reference (human) translation:



Syntactic AST Match

[[('d', 7, 'comesFrom', [], [])],
[('d', 16, 'comesFrom', ['d'], [7])],
[('d', 24, 'comesFrom', ['d'], [7])]]

Machine translation:

```
public static int Sign ( double d )
{
    return ((int)((d == 0)? 0:(d < 0)))?
    -1: 1;
}
```

Reference (human) translation:

```
public static short Sign ( double c )
{
    return (short)((c == 0)? 0:(c < 0)?
    -1: 1);
}
```

Semantic Data-flow Match

$$\text{CodeBLEU} = \alpha \cdot \text{N-Gram Match (BLEU)} + \beta \cdot \text{Weighted N-Gram Match} + \gamma \cdot \text{Syntactic AST Match} + \delta \cdot \text{Semantic Data-flow Match}$$